Output, income and capital flows

How do real resources become income, saving and financial claims?

The seven-class route

Class	Question
1. Accounting and flows	Who produces, spends and finances?
2. Money, inflation and activity	How does policy affect demand?
3. Bonds and yield curves	How are future payments priced?
4. Foreign exchange	How are currencies priced?
5. Open-economy policy	Which adjustment is possible?
6. Crises and the euro	How does financial fragility spread?
7. Debt and macro trades	How do fiscal and market risks interact?

LECTURE 1

Production and income

Why can we measure the same economy in three ways?

A loaf of bread



Invented bakery economy. The farmer, mill and bakery each add value.

Value added along the chain

Producer	Sales	Bought inputs	Value added
Farmer	10	0	10
Mill	25	10	15
Bakery	60	25	35
Total			60

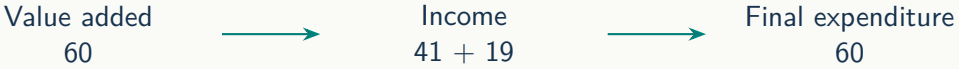
One period, one currency, no taxes or imports. Bread is the final good.

Output generates income

Producer	Wages	Operating surplus	Value added
Farmer	6	4	10
Mill	10	5	15
Bakery	25	10	35
Total	41	19	60

Each unit of value added accrues to labour or the business owners.

Three measurements of one flow



Production, income and expenditure count the same activity.

GDP counts current domestic production

Final goods and services produced inside the economy.

Investment includes new capital and changes in inventories.

Purchases of existing shares or old houses are asset trades.

Nominal output and real output

$$GDP_t^{\text{nominal}} = Q_t Y_t.$$

Y_t: real output. Q_t: the GDP price deflator.

Nominal growth combines more production and higher prices.

Growth rates separate prices and quantities

$$1 + g_{\text{nominal}} = (1 + g_{\text{real}})(1 + \pi_{\text{GDP}}).$$

$$1.03 \times 1.02 - 1 = 5.06\%.$$

Hypothetical 3% real growth and 2% GDP-deflator inflation.

Reading the GDP calendar

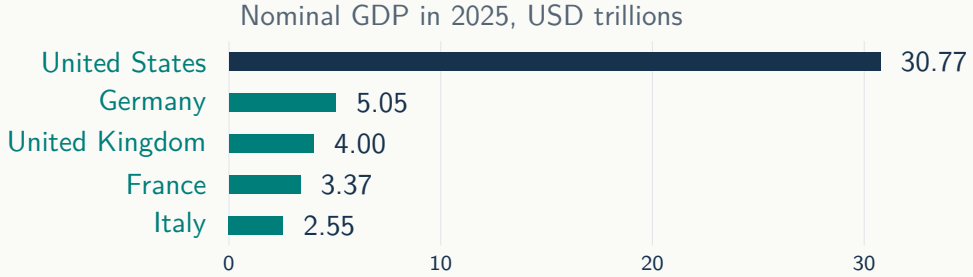
The US and euro-area countries measure GDP by quarter. Early estimates are revised as more data arrive.

Three US estimates of one quarter arrive in successive releases. The UK also publishes monthly real GDP.

The Fed has eight scheduled policy meetings a year. New data can change rate expectations between meetings.

BEA releases · ONS monthly GDP · FOMC calendar. Snapshot: 16 Oct 2026.

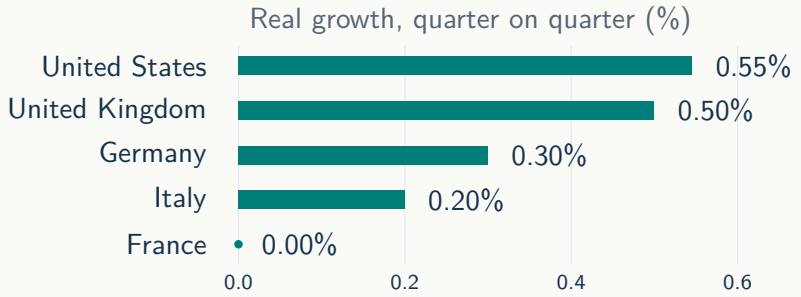
GDP levels: the size of an economy



Annual production at current prices, converted at market exchange rates. USD values also move with currencies; this is not a living-standards comparison.

Trading Economics / World Bank: explore the charts. 2025 data; accessed 16 Oct 2026.

Real GDP growth in Q2 2026



Release date

30 Sep 2026

30 Sep 2026

25 Aug 2026

5 Oct 2026

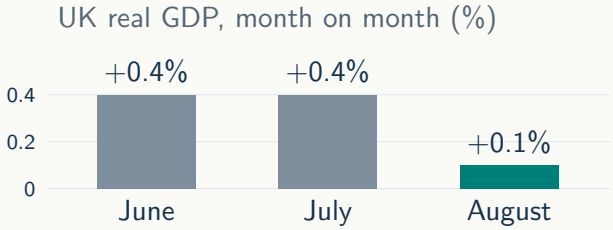
28 Aug 2026

$$g_{\mathrm{US,q/q}} = 100\left[(1.022)^{1/4} - 1\right] \approx 0.55\%.$$

US headline: 2.2% annualized, converted above. All bars are real, seasonally adjusted, nonannualized growth; European data also calendar adjusted.

The latest monthly GDP update

UK release: 15 Oct 2026. Reference month: August 2026.



June–August average GDP: +0.7% versus March–May.

ONS: explore charts and data. Revised estimates.

Italy, 16 Oct: September CPI: +4.2% y/y. Latest GDP (5 Oct): Q2 +0.2% q/q.

GDP news and market expectations

Italy, 30 July 2026: Q2 GDP $+0.2\%$, versus $+0.1\%$ consensus.

$$\underbrace{0.2\% - 0.1\%}_{\text{actual minus consensus}} = +0.1 \text{ percentage points.}$$

Stronger demand may lift profits. It may also raise expected interest rates, lowering bond prices.

Exercise: why might shares fall after good GDP news?

Italy GDP: chart and actual / consensus calendar. A surprise alone does not fix market direction.

LECTURE 1

Accounting and equilibrium

Which equalities always hold, and which need a model?

Expenditure in a closed economy

$$\boxed{Y = C + I + G.}$$

C: private consumption. I: gross domestic investment.

G: government purchases. No international trade yet.

Unsold bread still has an accounting destination

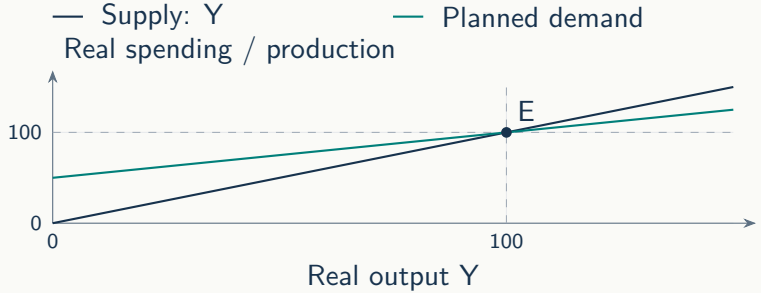
$$Y = 60, \qquad C = 50, \qquad I_{\text{inventory}} = 10.$$

Production is 60. Customers spend 50.

The bakery holds 10 of newly produced unsold bread.

Equilibrium adds planned spending

$$Y = C^{\text{plan}} + I^{\text{plan}} + G.$$



At E, planned spending equals output. Prices are fixed in this model.

Potential output and the long run

$$\boxed{Y^* = AK^{\alpha}(L^*)^{1-\alpha}, \qquad 0 < \alpha < 1.}$$

A: productivity. K: capital. L^*: labour at normal employment.

α: output elasticity of capital in this Cobb–Douglas model.

Potential is sustainable output. The long-run benchmark has $Y = Y^*$.

The output gap connects activity to rates

$$\text{Output gap} = 100\,\frac{Y - Y^*}{Y^*}.$$

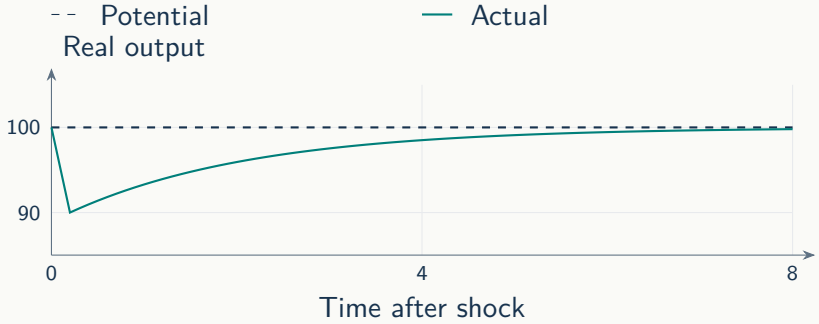
$$Y = 95, \quad Y^* = 100 \quad \Rightarrow \quad \text{gap} = -5\%.$$

Below potential: spare capacity can reduce inflation pressure.

Above potential: stretched resources can increase inflation pressure.

This helps interpret central-bank decisions. The next plot shows a gap closing.

A recession can be an equilibrium with slack



Hypothetical demand shock. The closing gap is a modelled adjustment.

Saving and investment in a closed economy

$$S \equiv Y - C - G.$$

$$Y = C + I + G \quad \Rightarrow \quad \boxed{S = I.}$$

Saving is income not consumed by households or government.

Private and public saving

$$S_p = Y - T - C, \qquad S_g = T - G.$$
$$S = S_p + S_g.$$

T: taxes less current transfers.

A public deficit may be matched by private saving or investment changes.

LECTURE 1

The rest of the world

How do trade, investment income and transfers connect?

Domestic expenditure and trade

$$\boxed{Y = C + I + G + X - M.}$$

X: exports. M: imports. $NX \equiv X - M$.

Imports are removed because spending categories include foreign output.

An imported machine

$$\Delta I = 10, \qquad \Delta M = 10.$$
$$\Delta Y = \Delta I - \Delta M = 0.$$

No extra domestic production at the purchase date in this stripped-down example.

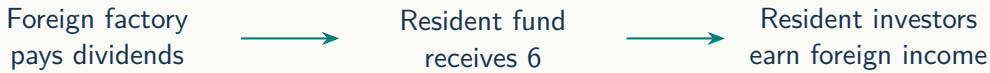
From domestic production to residents' income

$$GNI = Y + NPI.$$

NPI: net primary income received from abroad.

Think interest, dividends, reinvested earnings and cross-border labour income.

Primary income: an ownership example



Income received 6 − income paid abroad 2 = $NPI = 4$.

Invented resident/nonresident transactions, in one currency.

Current transfers add disposable income

$$GNDI = Y + NPI + NSI.$$

NSI: net secondary income from abroad.

A resident abroad sending a family remittance is a concrete example.

The current account is our net current flow from abroad

It adds what residents receive from the rest of the world, less what they pay, for trade, income and current transfers.

$$\boxed{CA = (X - M) + NPI + NSI.}$$

A deficit means current payments exceed receipts. The gap needs external financing.

This is why the current account connects trade to saving, investment and funding needs.

Saving finances investment at home or abroad

$$S \equiv GNDI - C - G.$$

$$GNDI = C + I + G + CA.$$

$$\boxed{S - I = CA.}$$

A deficit can finance productive investment

Economy	Saving	Investment	CA
New investment opportunity	20	30	-10
High consumption, same saving gap	10	20	-10

The same current-account deficit can reflect different economic stories.

LECTURE 1

US borrowing and dollar privilege

Why does the world keep financing the United States?

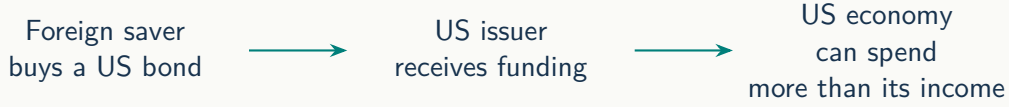
The financial counterpart

$$CA + KA = FA \quad \text{when measured without errors.}$$

KA: capital-account balance.

FA: net acquisition of foreign assets minus net incurrence of liabilities.

Foreign investors help finance US spending



$$CA_{\mathrm{US}} = -\$246.0 \text{ billion in Q2 2026.}$$

BEA, release 24 September 2026. The diagram illustrates one financing channel.

Net international investment position

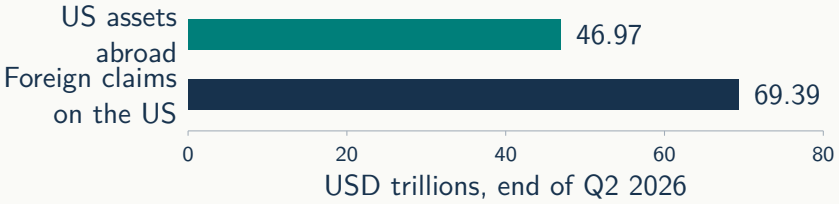
$$\boxed{NIIP = \text{assets held abroad} - \text{foreign claims on us.}}$$

The current account is a flow over a period. NIIP is a balance sheet at a date.

A negative NIIP means foreign claims exceed residents' foreign assets.

Claims include equity and direct investment as well as debt.

The US external balance sheet is large on both sides



$$NIIP = 46.97 - 69.39 = -\$22.42 \text{ trillion.}$$

BEA, end of Q2 2026. These positions include corporate equity and direct investment.

What is the US “exorbitant privilege”?

Investors want dollar assets for reserves, payments and collateral.

They may accept a lower yield for those services, reducing US funding costs.

The US can borrow in dollars while owning riskier businesses abroad.

Dollar funding advantages and returns on the foreign portfolio both matter.

Negative NIIP can coexist with positive investment income

Hypothetical external portfolio	Position	Income yield	Annual income
Foreign assets owned	100	6%	+6
Liabilities to foreigners	150	2%	-3

$$NIIP = 100 - 150 = -50, \qquad \text{net investment income} = 6 - 3 = +3.$$

Illustrative income yields, excluding capital gains. These are not current US observations.

US asset demand connects yields and the dollar

Conditional change	Other inputs fixed
Fewer Treasury buyers	Lower prices, higher yields
Less demand for dollar assets	Pressure on the dollar
A weaker dollar	Higher dollar value of unhedged foreign assets

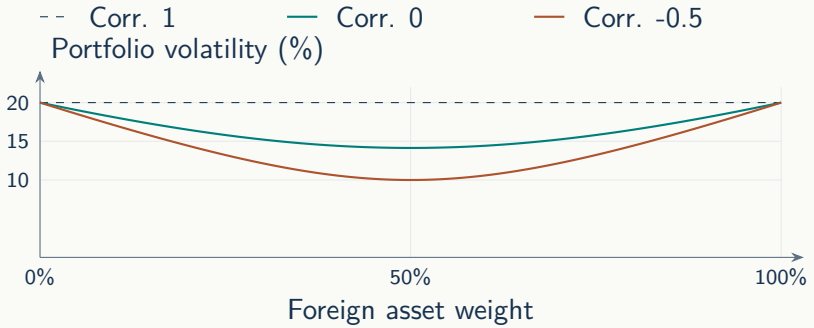
These are conditional mechanisms. NIIP alone gives no market forecast.

Risk sharing is a gain from international finance

Two equally likely states	Home harvest	Foreign harvest
State A	120	80
State B	80	120

Half-ownership of each harvest pays 100 in either state.

Diversification depends on correlation



Equal standalone volatility of 20%. Hypothetical assets and correlations.

More about portfolio diversification in the future.

What a macro release can and cannot tell a trader

GDP describes production relative to expectations and capacity.

The current account describes net external funding needs.

The US example shows why asset demand, income yields and portfolio risk matter.

Counterfactual exercise: extra public consumption

Government consumption rises by 2% of GDP.

Taxes, disposable income and investment initially stay fixed.

Predict the change in national saving and the current account.

The accounting answer and the economic question

$$\Delta S = -2\%, \qquad \Delta CA = -2\% \text{ of GDP}.$$

If output or private saving responds, those numbers change.

Next lecture supplies a model of demand and rates.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.